

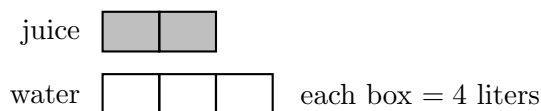
Ratios with Tape Diagrams and Double Number Lines

Tape diagrams for sharing and differences, and double number lines for ratio problems

Directions:

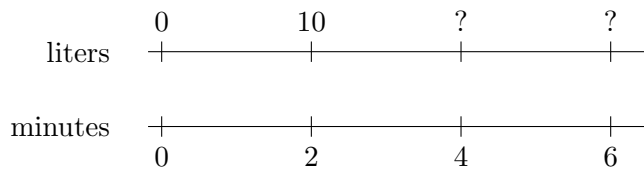
- A **tape diagram** shows a ratio as rows of equal boxes. Find what one box is worth first, then count boxes.
- A **double number line** shows two quantities that grow together. Matching marks always show the same ratio.
- Show your steps in the work space, and include units in your thinking.

1. The tape diagram shows the juice and water in a fruit punch. Every box holds the same amount, and each box stands for 4 liters. How many liters of juice are in the punch?



Answer: _____ L

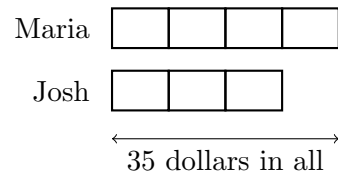
2. A hose fills 10 liters of water every 2 minutes. Complete the double number line, then write the two missing liter values.



(a) liters at 4 minutes: _____

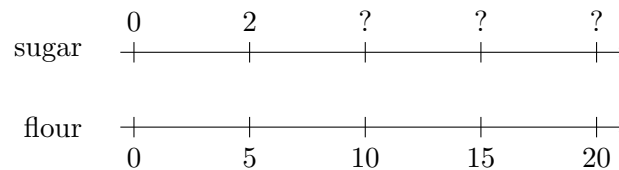
(b) liters at 6 minutes: _____

- 3.** Maria and Josh share 35 dollars in the ratio 4:3, as the tape diagram shows. How many dollars does Maria receive?



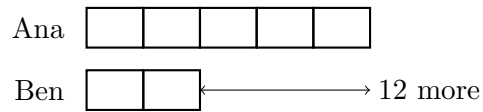
Answer: _____

- 4.** A muffin recipe uses 2 cups of sugar for every 5 cups of flour. Complete the sugar row of the double number line.



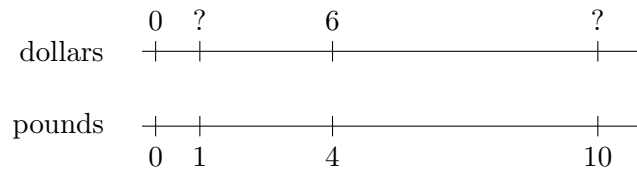
cups of sugar for 10 cups of flour: _____ for 15 cups: _____ for 20 cups: _____

- 5.** Ana has 12 more stickers than Ben, and the tape diagram shows their sticker counts in the ratio 5:2. How many stickers does Ana have? Hint: the extra boxes make up the difference of 12.



Answer: _____

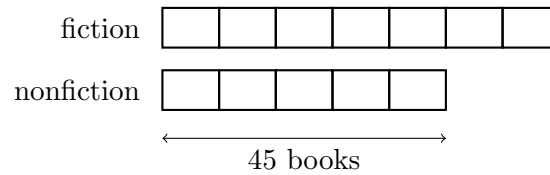
- 6.** Apples cost 6 dollars for 4 pounds. Use the double number line to find each missing money value.



(a) dollars for 1 pound: _____

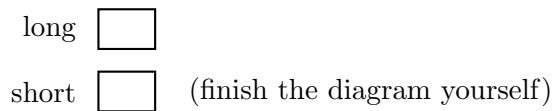
(b) dollars for 10 pounds: _____

7. A library shelf holds fiction and nonfiction books in the ratio 7:5. The tape diagram shows that the 5 nonfiction boxes hold 45 books altogether. How many fiction books are on the shelf?



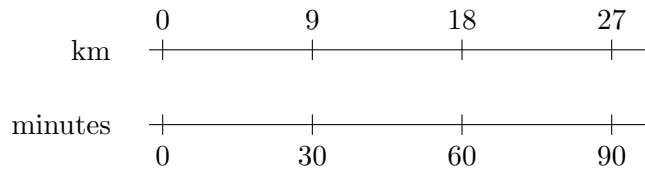
Answer: _____

8. A 91-centimeter rope is cut into two pieces whose lengths have the ratio 9:4. Finish drawing the tape diagram in the work space, then find how many centimeters **longer** the long piece is than the short piece.



Answer: _____ cm

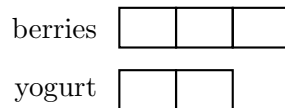
9. A cyclist rides 27 kilometers in 90 minutes at a steady pace. The double number line shows her progress. Use it to answer both questions.



(a) kilometers ridden in 10 minutes: _____

(b) kilometers ridden in 50 minutes: _____

10. A smoothie recipe mixes berries and yogurt in the ratio 3:2, shown by the blank tape diagram below. A big batch uses 12 cups of berries. How many cups of yogurt does it use? Then explain, in one sentence in the work space, how the tape diagram helped you find the answer.



Answer: _____